

Interest Rate Models for Quants

PnL, Hedging, and Model Choice Across the Yield Curve
Why Desks Choose Hull–White, LMM, SABR — and When They Break

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Module 2: Why Simple Models Fail—The Naïve Approach

2.1 Three Deadly Naïve Ideas

"Simple Models Assume Markets Are Stupid" **Memory:** If it were this easy, everyone would arb it out of existence.

Naïve Idea	Why It Fails (Economically)	Rates Model Fix
Deterministic Forward Curve	Predicts exact future rates $\rightarrow$ infinite arbitrage if wrong. No uncertainty.	Add stochastic shocks: $dr_t = \mu dt + \sigma dW_t$
Flat Curve Dynamics	Curve twists and kinks $\rightarrow$ parallel hedge fails.	Use key-rate DV01 or 2-factor models
No Mean Reversion	Rates drift to infinity or negative $\rightarrow$ nonsense.	Add $\kappa(\theta - r_t)$ term
One-Factor Model	Curve shape ignored $\rightarrow$ butterfly risk unhedged.	Multi-factor (2F-HW, LMM)
Constant Volatility	Vol smiles, clusters $\rightarrow$ swaption mispricing.	Stoch vol (SABR, Heston) or time-dependent

Real-World Failure: - **Naïve:** "10Y rate will be 5- **Reality:** Could be 2

Naïve Failure"Simple models create arbitrage and miss curve dynamics" $y(t) \neq \text{constant}$ "Interview: 'I never use flat curve assumptions. The curve moves in three dimensions: level, slope, curvature.'"

2.2 Why Point Forecasts Fail in Rates

Fallacy: "The 10Y rate will be 6% in 2 years."

Reality: Pricing depends on the **distribution**, not the point estimate.

Core Formula

$$\text{Derivative Price} = \mathbb{E}^Q[\text{Payoff}]$$

Two forecasts with the same mean but different variances produce **very different option prices**.

"Rates Models Price Uncertainty, Not Opinions" **Memory:** Mean forecasts do not price convexity. **Interview:** "In rates, the variance matters more than the mean for options."

2.3 Hedging a 10Y Swap with 2Y Rate: The Mismatch

Trade: Pay fixed 10Y, receive floating. Hedge with 2Y payer swap.

Naïve hedge ratio:

Core Formula

$$N_{2Y} = \frac{DV01_{10Y}}{DV01_{2Y}} \times \text{Notional}_{10Y}$$

What goes wrong: - Curve steepens: 2Y rate falls, 10Y rate rises $\rightarrow$ lose on both legs - Curve flattens: opposite $\rightarrow$ win - **You're short curve steepness risk, not hedged**

Correct hedge: - Key-rate DV01: hedge each bucket (2Y, 5Y, 10Y) separately - Or use 2-factor model that captures slope

Core Formula

$$\frac{\partial P}{\partial r_t} = -B(t, T)P(t, T)$$

Convexity:

Core Formula

$$\frac{\partial^2 P}{\partial r_t^2} = B(t, T)^2 P(t, T)$$

"Vasicek Greeks Are Closed-Form and Beautiful" **Numerical Example:** $r_t = 3\%$, $\kappa = 0.5$, $\theta = 4\%$, $\sigma = 2\%$, $T - t = 10Y$:

$$B = \frac{1}{0.5}(1 - e^{-5}) = 1.986$$

$$P = e^{-1.986 \times 0.03} = 0.942 \quad \$94.2\% \text{ of par}$$

$$DV01 = 1.986 \times 0.942 \times 0.0001 = 0.000187 \quad \$187/\text{MM per bp}$$

3.9 HW Calibration Step-by-Step

Step 1: Extract market discount factors

Core Formula

$$P^{\text{mkt}}(0, T_i) \quad \text{for all maturities } T_i$$

Step 2: Solve for $\theta(t)$ analytically

Core Formula

$$\theta(t) = f(0, t) + \frac{1}{\kappa} \frac{\partial f(0, t)}{\partial t} + \frac{\sigma^2}{2\kappa^2} (1 - e^{-2\kappa t})$$

where $f(0, t)$ is instantaneous forward rate.

Step 3: Calibrate κ, σ to swaption vols

Core Formula

$$\min_{\kappa, \sigma} \sum_{i,j} (\sigma_{\text{model}}(T_i, T_j) - \sigma_{\text{mkt}}(T_i, T_j))^2$$

HW Calibration "Fit $\theta(t)$ to curve, fit (κ, σ) to vols" Two-stage optimization, stable if done in order "Interview: 'I first fit the yield curve perfectly with $\theta(t)$, then calibrate κ and σ to ATM swaptions. This prevents overfitting.'"

3.9 CIR Option Pricing (Brief Sketch)

Bond price: Non-central chi-square distribution

Core Formula

$$P(t, T) = \chi^2(\cdot; \text{df}, \text{ncp})$$

Call on zero-coupon bond:

Core Formula

$$C(t, T, S, K) = P(t, S)\chi^2(a; d, \lambda_1) - KP(t, T)\chi^2(a; d, \lambda_2)$$

"CIR Gives Closed-Form But Messy Formulas" **Memory:** CIR bond options exist but involve chi-square CDFs.
Implementation: Use numerical libraries for chi-square.

3.10 Quick Interview Cheat Sheet — Module 3

Module 3: Short-Rate = Speed vs Accuracy

Core Concept: Choose model by **trade-off**: HW (fast), CIR (positive), G2++ (curve shape).

Must-Know Formulas:

- **Vasicek/HW:** $dr = \kappa(\theta - r)dt + \sigma dW$
- **Half-life:** $\ln(2)/\kappa$
- **HW Bond:** $P(t, T) = A(t, T)e^{-B(t, T)r_t}$

Interview Triggers:

- "HW vs Vasicek?" → You say: "HW adds time-varying $\theta(t)$ to fit today's curve exactly."
- "CIR positivity?" → You say: " $2\kappa\theta > \sigma^2$ (Feller). Vol collapses near zero."
- "G2++?" → You say: "Two factors capture level + slope twist for Bermudans."

Model Selection:

- **Bermudan:** HW-2F (fast exercise)
- **Credit intensity:** CIR (positive only)
- **Quick PV:** HW-1F

Common Trap: "Don't calibrate κ from flat vol matrix—it's unidentified."

Memory Aid: "Vasicek = rubber band (no floor). CIR = rubber band + floor. HW = Vasicek glued to today's curve."

Interviewer Is Testing: Can you calibrate **two-stage** (curve then vol)? Do you know when each model breaks?